Case study

At Adventure Works, the warehouse team wants to analyze the current inventory range and stock levels. They've provided you with a dataset but have notified you that there are currently issues with their inventory system. As a data analyst, you know the importance of consistent and accurate data. Therefore, you want to evaluate and inspect the data that the company provided you with.

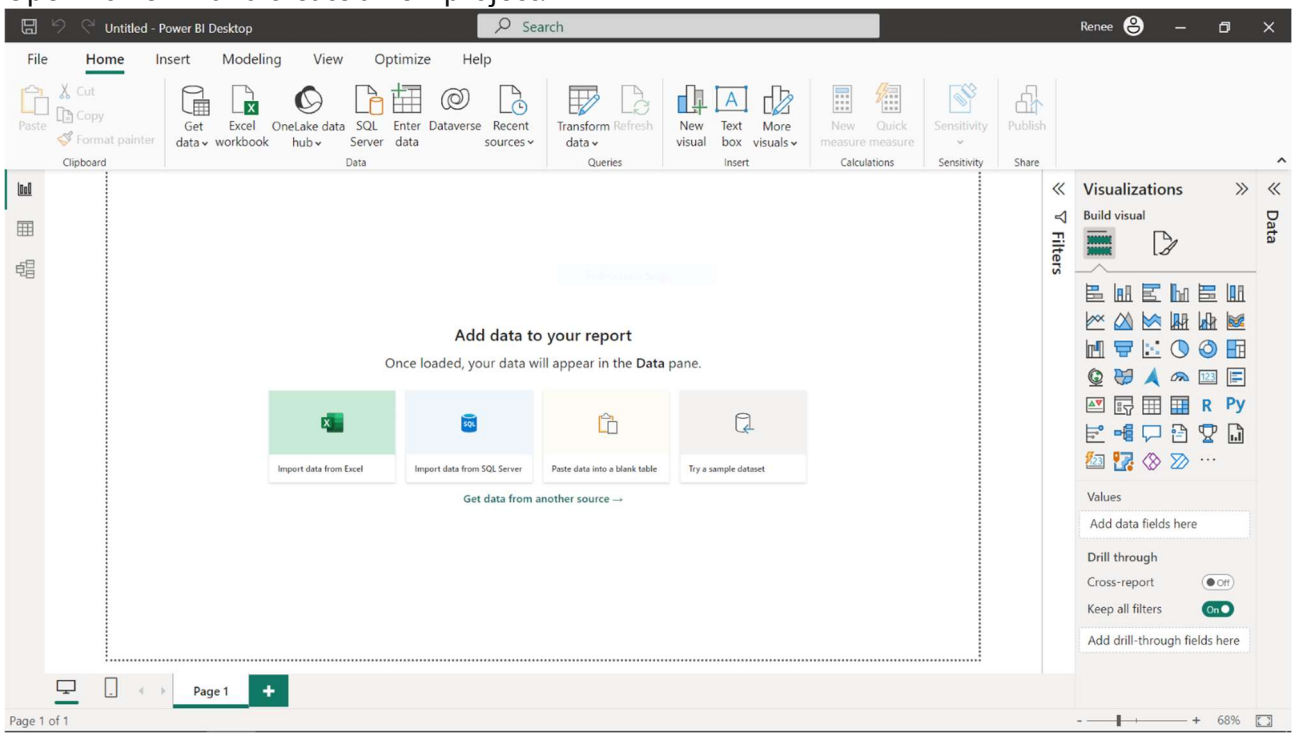
Instructions

Step 1: Connect to the data source

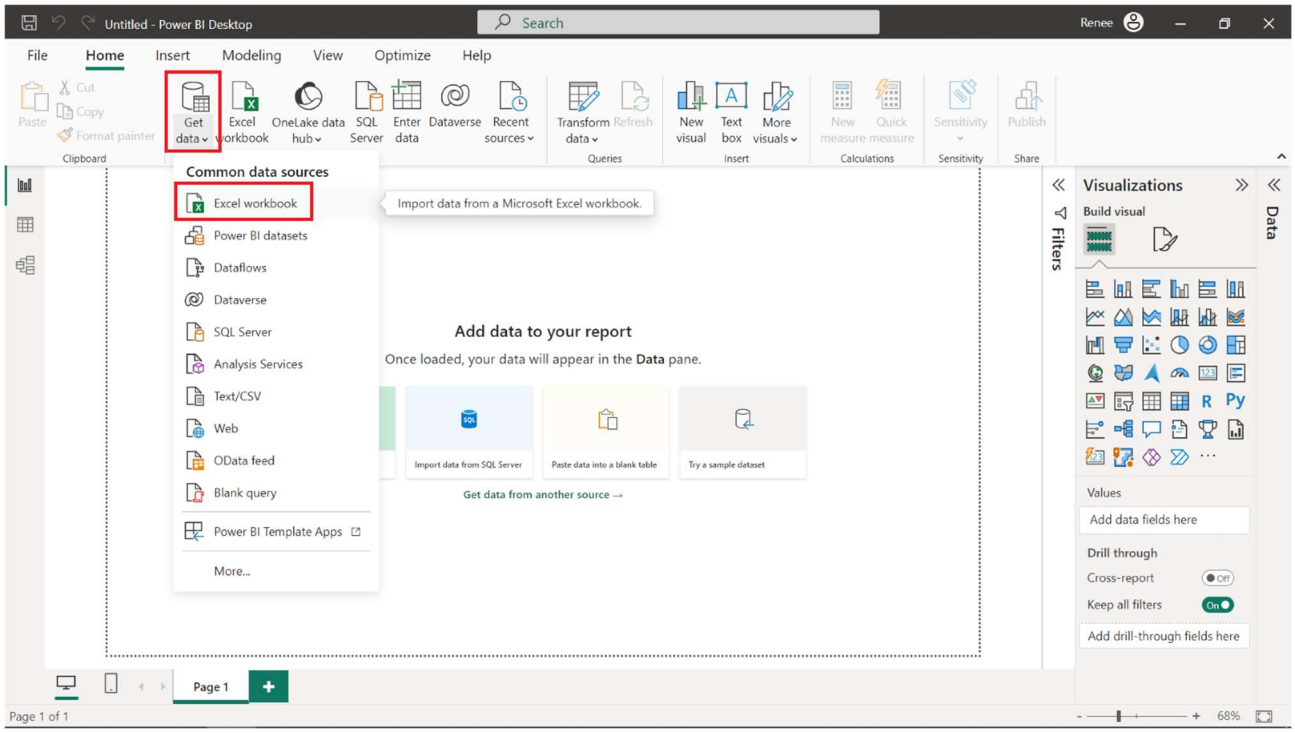
1. Download the *AdventureWorksAnalysis.xlsx* dataset.

module-1_preparing-data/02_profiling-and-cleaning-data / exercises/02_AnalyzeAdventureWorksAnalysis /datasets/AUbmCltlSrmTaHyhHhHZYw_bb22b09d82ac4157825581728c968fe1_AdventureWorksSales.xlsx

2. Open Power BI and create a new project.

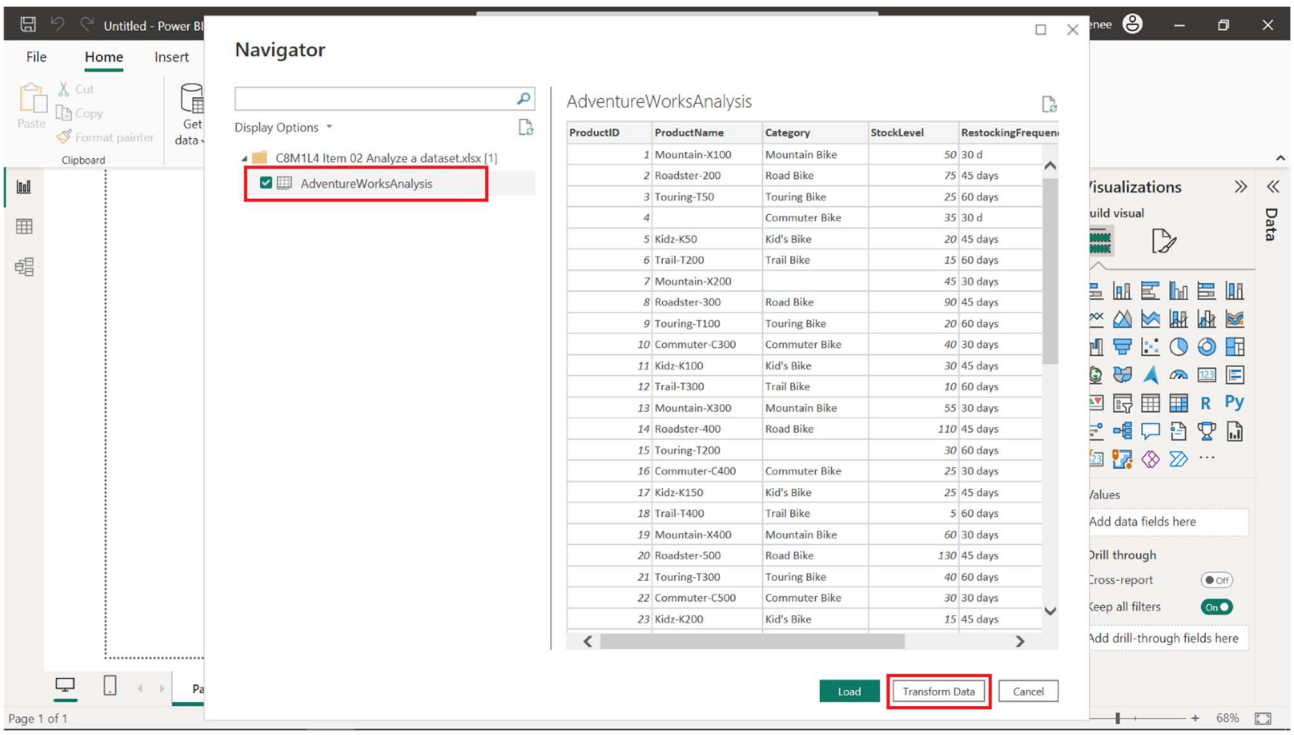


3. On the **Home** tab of the ribbon, select **Get Data** and choose **Excel workbook**.



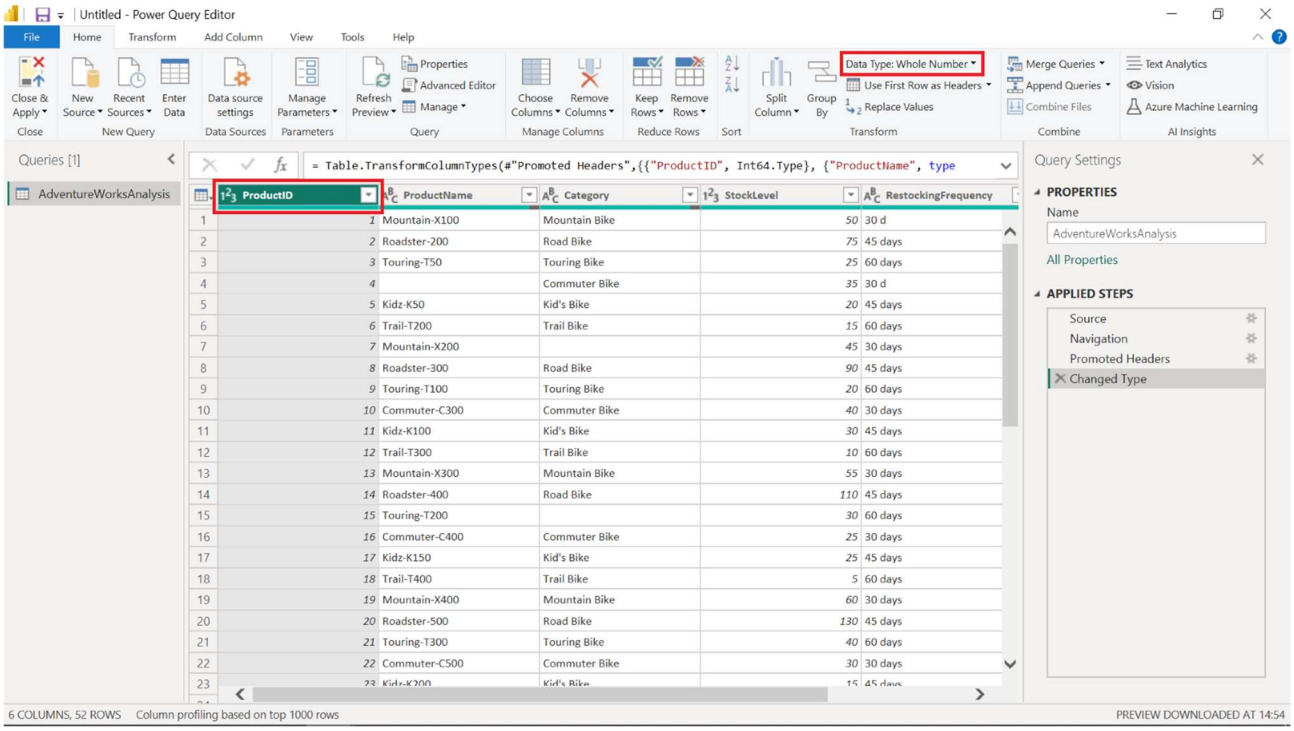
4. When the file browser opens, select the *AdventureWorksAnalysis.xlsx* file.

5. In the **Navigator** window, select the *AdventureWorksAnalysis* sheet, followed by **Transform Data.**

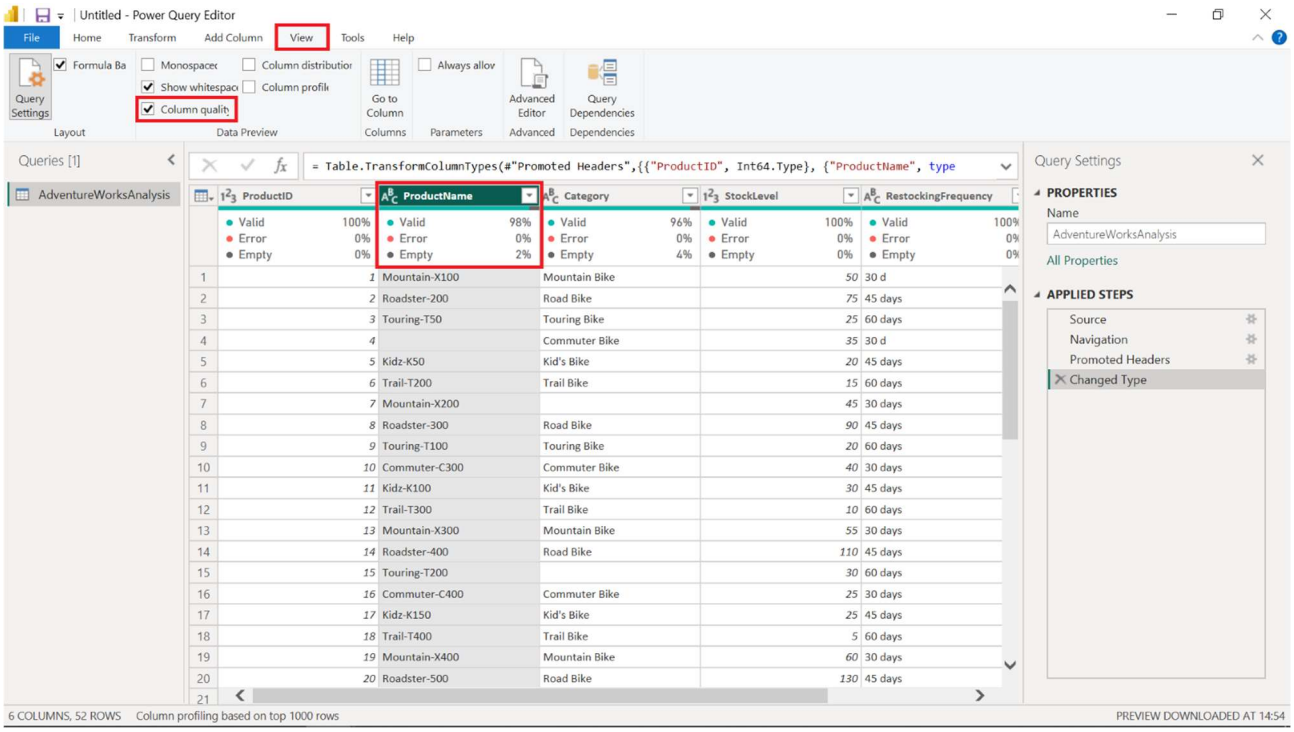


Step 2: Inspect the data quality in the Power Query Editor

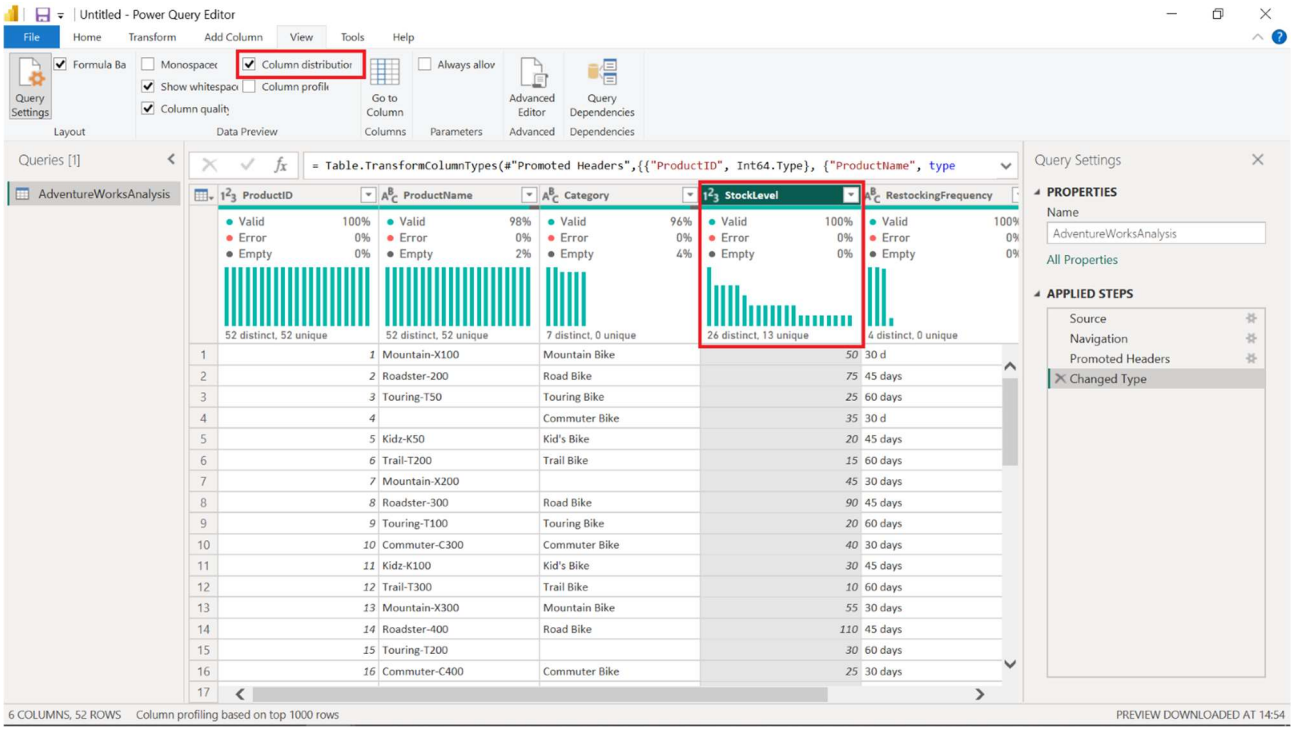
1. When the Power Query Editor opens, select the **ProductID** column and ensure it is set to the **Whole Number** data type.



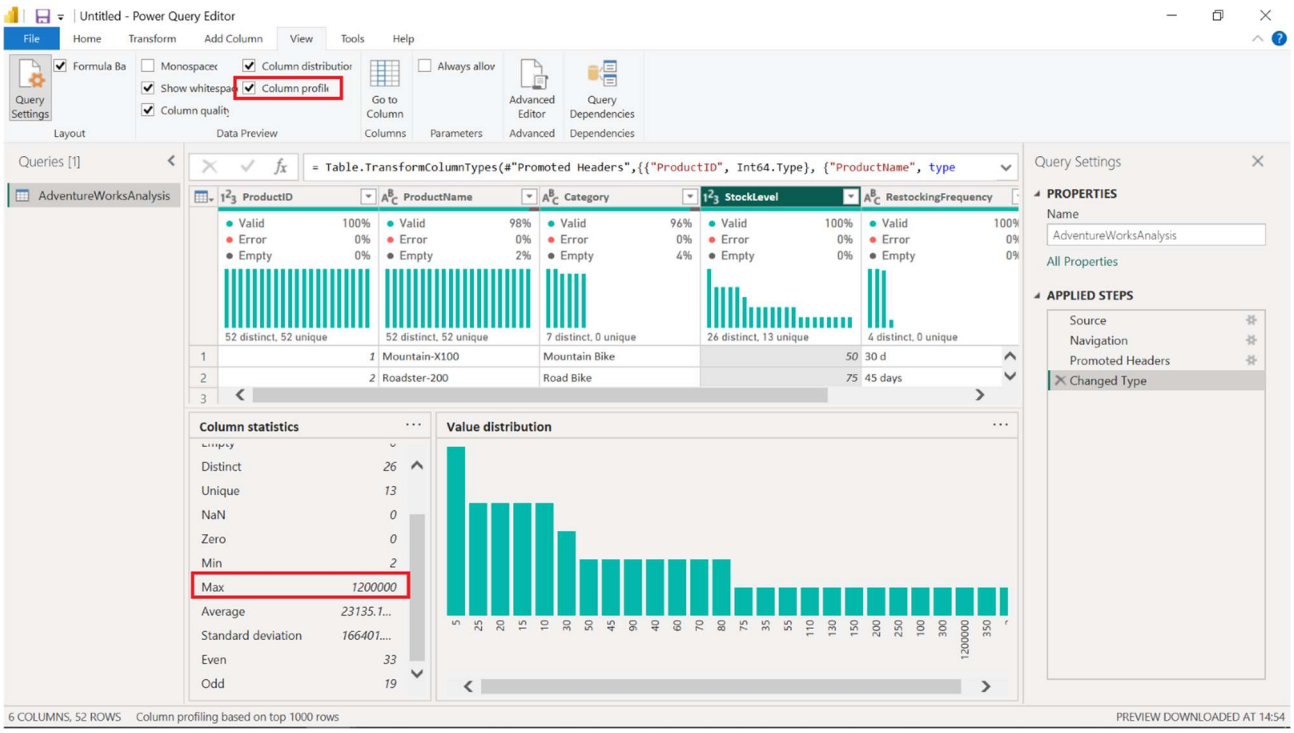
1. Then, go to the **View** tab in the ribbon menu. Enable **Column quality** and inspect the quality values for the **ProductName** column. Note that 98% of values in the column are valid and that 2% of values are empty.



1. Next, enable **Column distribution**. Inspect the distribution for the **StockLevel** column. Note that there are 26 distinct values and 13 unique values. This means that there are 26 values within this column for the dataset and that 13 of them only occur once.



1. After that, enable **Column profile**. Inspect the profile of the **StockLevel** column. Note the statistical information in the profile and that the maximum value is 1,200,000. The Adventure Works warehouse cannot store this number of items, indicating an anomaly in the data.



1. In the top-left corner of the Power Query Editor, select **Close & Apply**. Save the project to your computer.

Conclusion

Congratulations! You have successfully inspected the data quality using Power Query and identified irregularities in the data. As a data analyst, monitoring data quality will help you ensure the accuracy and consistency of data as you proceed through the data analysis process.